Part II: Simulator

AstroConcepts Multi-Label Classification Pipeline

This lightweight, runnable Python simulator demonstrates the project's input-to-output classification workflow. It takes a paper title and abstract and returns ranked astronomy concept labels, heuristic scores, prediction sources, and simulated interpretability hints. It uses an eight-paper synthetic corpus and a keyword-rule + TF-IDF/logistic-regression hybrid rather than the full AstroConcepts dataset or a trained astroBERT model.

1. Scope and Important Limitations

- The synthetic corpus contains 22 distinct demonstration labels. It does not reproduce the full project label space or validate official UAT identifiers.
- Scores combine keyword matches and logistic-regression outputs; they are not calibrated confidence probabilities.
- Interpretability hints use keyword-token overlap. They are illustrative only and do not implement LIME or SHAP.
- This demonstration does not train an RNN, Transformer, or autoencoder, and does not perform held-out evaluation. It simulates the classification interface, not every project experiment.

2. How to Run

```
pip install scikit-learn numpy
python simulator.py
```

Save the supplied script as simulator.py. Running it trains the small baseline and prints predictions for three demonstration papers. No dataset or pretrained-model download is needed.

3. Pipeline Design

1. Combine title and abstract into a single input string.
2. Encode the synthetic training labels with MultiLabelBinarizer and text with unigram/bigram TF-IDF.
3. Train one-vs-rest logistic regression with balanced class weights.
4. Compute keyword-rule scores and combine them with logistic-regression outputs: min(1, max(rule, LR) + agreement bonus). The bonus is 0.1 when the rule score is positive and the LR output exceeds 0.2.
5. Sort deterministically by descending fused score, then label name for ties; retain up to six predictions at the default threshold of 0.35.
6. Display matched-token hints for each retained concept.

4. Complete Runnable Python Simulator

```
"""AstroConcepts Multi-Label Classification Simulator.
Run: python simulator.py
Scores are heuristic, not calibrated confidence probabilities.
Evidence is keyword-based, not actual LIME or SHAP attribution.
"""
import re
```

```
import numpy as np
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.multiclass import OneVsRestClassifier
from sklearn.preprocessing import MultiLabelBinarizer

CORPUS = [
    ("Galaxy evolution in the local universe",
     "We study the star formation history of nearby galaxies using spectroscopy. "
     "The galaxy evolution is driven by mergers and active galactic nuclei feedback.",
     ["Galaxy evolution", "Star formation", "Spectroscopy"]),
    ("Exoplanet atmospheres with JWST",
     "Transmission spectroscopy of exoplanet atmospheres reveals water and methane. "
     "The exoplanet atmosphere is characterized by clouds and hazes.",
     ["Exoplanet atmospheres", "Exoplanets", "Spectroscopy"]),
    ("Neutron star mergers and gravitational waves",
     "We model the kilonova emission from neutron star mergers. "
     "Gravitational waves from binary neutron stars constrain the equation of state.",
     ["Neutron stars", "Gravitational waves", "Nuclear astrophysics"]),
    ("Active galactic nuclei variability",
     "We analyze the X-ray variability of active galactic nuclei. "
     "The accretion disk and corona drive the observed flux changes.",
     ["Active galactic nuclei", "Accretion", "X-ray astronomy"]),
    ("Star formation in molecular clouds",
     "We observe star formation in molecular clouds using radio interferometry. "
     "Dense cores collapse to form protostars and protoplanetary disks.",
     ["Star formation", "Molecular clouds", "Radio astronomy"]),
    ("Cosmology with the cosmic microwave background",
     "We constrain cosmological parameters from cosmic microwave background anisotropies. "
     "The power spectrum probes inflation and dark energy.",
     ["Cosmology", "Cosmic microwave background", "Dark energy"]),
    ("Solar flares and coronal mass ejections",
     "We study solar flares and coronal mass ejections using extreme ultraviolet imaging. "
     "Magnetic reconnection powers the solar activity.",
     ["Solar flares", "Coronal mass ejections", "Solar activity"]),
    ("Dark matter halos and galaxy clustering",
     "We use N-body simulations to study dark matter halos and galaxy clustering. "
     "Halo mass function constrains dark matter properties.",
     ["Dark matter", "Galaxy clustering", "N-body simulations"]),
]

KEYWORD_RULES = {
    "Galaxy evolution": ["galaxy evolution", "galaxy", "merger"],
    "Star formation": ["star formation", "protostar", "protostellar"],
    "Spectroscopy": ["spectroscopy", "spectroscopic", "spectra"],
    "Exoplanet atmospheres": ["exoplanet atmosphere", "transmission spectroscopy"],
    "Exoplanets": ["exoplanet", "planet"],
    "Neutron stars": ["neutron star", "kilonova"],
    "Gravitational waves": ["gravitational wave", "binary neutron star"],
    "Nuclear astrophysics": ["nuclear astrophysics", "equation of state"],
    "Active galactic nuclei": ["active galactic nuclei", "agn", "accretion disk"],
    "Accretion": ["accretion", "corona"],
    "X-ray astronomy": ["x-ray", "xray"],
    "Molecular clouds": ["molecular cloud", "dense core"],
    "Radio astronomy": ["radio interferometry", "radio"],
    "Cosmology": ["cosmology", "cosmological", "inflation"],
    "Cosmic microwave background": ["cosmic microwave background", "cmb"],
    "Dark energy": ["dark energy"],
    "Solar flares": ["solar flare", "flare"],
    "Coronal mass ejections": ["coronal mass ejection", "cme"],
    "Solar activity": ["solar activity", "magnetic reconnection"],
    "Dark matter": ["dark matter", "halo"],
    "Galaxy clustering": ["galaxy clustering"],
    "N-body simulations": ["n-body", "simulation"],
```

```
}

texts = [title + " " + abstract for title, abstract, _ in CORPUS]
labels = [items for _, _, items in CORPUS]
mlb = MultiLabelBinarizer()
Y = mlb.fit_transform(labels)
vectorizer = TfidfVectorizer(ngram_range=(1, 2), stop_words="english")
X = vectorizer.fit_transform(texts)
clf = OneVsRestClassifier(
    LogisticRegression(max_iter=1000, class_weight="balanced")
)
clf.fit(X, Y)

def _token_attribution(text, label, top_k=4):
    """Return keyword-overlap hints, not model-faithful explanations."""
    tokens = re.findall(r"[a-zA-Z\-]+", text.lower())
    hits = []
    for kw in KEYWORD_RULES.get(label, []):
        kw_tokens = set(re.findall(r"[a-zA-Z\-]+", kw.lower()))
        hits.extend(tok for tok in tokens if tok in kw_tokens)
    out = list(dict.fromkeys(hits))
    return out[:top_k] or ["(no strong single-token evidence)"]

def _rule_scores(text):
    low = text.lower()
    scores = {}
    for label, keywords in KEYWORD_RULES.items():
        count = sum(1 for kw in keywords if kw in low)
        if count:
            scores[label] = count / len(keywords)
    return scores

def predict(title, abstract, threshold=0.35, top_n=6):
    """Return (label, heuristic score, source, evidence tokens) tuples."""
    if not 0 <= threshold <= 1:
        raise ValueError("threshold must be between 0 and 1")
    if top_n < 1:
        raise ValueError("top_n must be positive")
    full = title + " " + abstract
    rule = _rule_scores(full)
    x = vectorizer.transform([full])
    probs = (clf.predict_proba(x)[0] if x.nnz
             else np.zeros(len(mlb.classes_)))
    lr = {label: float(prob) for label, prob in zip(mlb.classes_, probs)}
    fused = {}
    for label in sorted(set(rule) | set(lr)):
        r, prob = rule.get(label, 0.0), lr.get(label, 0.0)
        bonus = 0.1 if r > 0 and prob > 0.2 else 0.0
        fused[label] = round(min(1.0, max(r, prob) + bonus), 3)
    ranked = sorted(fused.items(), key=lambda item: (-item[1], item[0]))
    results = []
    for label, score in ranked:
        if score < threshold:
            continue
        sources = []
        if rule.get(label, 0) > 0:
            sources.append("rule")
        if lr.get(label, 0) > 0.2:
            sources.append("LR")
        results.append((label, score, "+".join(sources) or "fused",
                        _token_attribution(full, label, top_k=3)))
        if len(results) >= top_n:
            break
    return results
```

```
def run_simulator(title, abstract, threshold=0.35, top_n=6):
    print("=" * 90)
    print("INPUT")
    print("Title   :", title)
    print("Abstract:", abstract)
    print("\nOUTPUT - Predicted UAT Concepts (multi-label)")
    preds = predict(title, abstract, threshold, top_n)
    if not preds:
        print("No concept above threshold.")
        return
    print(f"{'Rank':<5}{'UAT Concept':<32}{'Score':<8}{'Source':<14}Evidence (simulated)")
    print("-" * 90)
    for i, (label, score, source, evidence) in enumerate(preds, 1):
        print(f"{i:<5}{label:<32}{score:<8.3f}{source:<14}{', '.join(evidence)}")
    print(f"Total predicted labels: {len(preds)} | Threshold: {threshold}")
    print("Scores are heuristic, not calibrated confidence probabilities.")
    print("Evidence is keyword overlap, not actual LIME/SHAP attribution.")

DEMO_CASES = [
    ("JWST transmission spectroscopy of a hot Jupiter",
     "We present JWST transmission spectroscopy of the exoplanet atmosphere of a hot Jupiter. "
     "The spectrum reveals water, carbon dioxide, and clouds. This exoplanet is a benchmark "
     "for atmospheric characterization and spectroscopy."),
    ("Star formation and feedback in a nearby galaxy",
     "Using radio interferometry we map molecular clouds and star formation in a nearby galaxy. "
     "Stellar feedback drives galaxy evolution and regulates star formation."),
    ("A new cosmological constraint from CMB lensing",
     "We analyze cosmic microwave background lensing to constrain cosmology and dark energy. "
     "The power spectrum is consistent with the standard cosmological model."),
]

if __name__ == "__main__":
    for title, abstract in DEMO_CASES:
        run_simulator(title, abstract)
        print()
```

5. Example Output from an Executed Run

The following output was captured by running the included script. Unlike a hand-written illustration, it reflects the implemented fusion rule and threshold. Numeric values may differ across library versions.

```
==========================================================================================
INPUT
Title   : JWST transmission spectroscopy of a hot Jupiter
Abstract: We present JWST transmission spectroscopy of the exoplanet atmosphere of a hot Jupiter. The spectrum
reveals water, carbon dioxide, and clouds. This exoplanet is a benchmark for atmospheric characterization and
spectroscopy.

OUTPUT - Predicted UAT Concepts (multi-label)
Rank UAT Concept                     Score   Source        Evidence (simulated)
------------------------------------------------------------------------------------------
1    Exoplanet atmospheres           1.000   rule+LR       exoplanet, atmosphere, transmission
2    Exoplanets                      1.000   rule+LR       exoplanet
3    Spectroscopy                    0.661   rule+LR       spectroscopy
4    Star formation                  0.406   LR            (no strong single-token evidence)
5    Galaxy evolution                0.350   LR            (no strong single-token evidence)
Total predicted labels: 5 | Threshold: 0.35
Scores are heuristic, not calibrated confidence probabilities.
Evidence is keyword overlap, not actual LIME/SHAP attribution.

==========================================================================================
INPUT
Title   : Star formation and feedback in a nearby galaxy
```

```
Abstract: Using radio interferometry we map molecular clouds and star formation in a nearby galaxy. Stellar
feedback drives galaxy evolution and regulates star formation.

OUTPUT - Predicted UAT Concepts (multi-label)
Rank UAT Concept                    Score   Source        Evidence (simulated)
-------------------------------------------------------------------------------------------
1    Radio astronomy                1.000   rule+LR       radio, interferometry
2    Galaxy evolution               0.767   rule+LR       galaxy, evolution
3    Star formation                 0.698   rule+LR       star, formation
4    Molecular clouds               0.600   rule+LR       molecular
5    Spectroscopy                   0.486   LR            (no strong single-token evidence)
Total predicted labels: 5 | Threshold: 0.35
Scores are heuristic, not calibrated confidence probabilities.
Evidence is keyword overlap, not actual LIME/SHAP attribution.

===========================================================================================
INPUT
Title   : A new cosmological constraint from CMB lensing
Abstract: We analyze cosmic microwave background lensing to constrain cosmology and dark energy. The power
spectrum is consistent with the standard cosmological model.

OUTPUT - Predicted UAT Concepts (multi-label)
Rank UAT Concept                    Score   Source        Evidence (simulated)
-------------------------------------------------------------------------------------------
1    Cosmic microwave background    1.000   rule+LR       cosmic, microwave, background
2    Dark energy                    1.000   rule+LR       dark, energy
3    Cosmology                      0.767   rule+LR       cosmology, cosmological
4    Spectroscopy                   0.385   LR            (no strong single-token evidence)
5    Star formation                 0.378   LR            (no strong single-token evidence)
6    Dark matter                    0.351   LR            dark
Total predicted labels: 6 | Threshold: 0.35
Scores are heuristic, not calibrated confidence probabilities.
Evidence is keyword overlap, not actual LIME/SHAP attribution.
```

6. How This Maps to the Actual Project

Simulator Component	Real Project Replacement
CORPUS (8 synthetic papers)	Use load_data() from utils.py to load the fixed AstroConcepts train/validation/test partitions.
22 demonstration labels and MultiLabelBinarizer	Use a fixed label-to-index mapping for the intended 2,367-concept label space; preserve the mapping across all splits.
TF-IDF + one-vs-rest logistic regression	Retain as a classical baseline; compare against LSTM/GRU and fine-tuned astroBERT.
KEYWORD_RULES	Keep as a separately evaluated rule-based baseline, or document any hybrid rules explicitly.
Heuristic score fusion and threshold 0.35	Select thresholds on validation data; report held-out classification and ranking metrics. Do not interpret fused scores as calibrated probabilities.
_token_attribution()	Replace with actual LIME/SHAP analysis of selected label outputs; document the explainer configuration.
Three demo cases	Use the fixed test set to report Macro-F1, Micro-F1, Precision@k, Recall@k, and head/torso/tail performance.
No latent-space module	Add the separate autoencoder experiment and 2D visualization from the proposal.
Local execution	Run lightweight work locally and reserve budgeted cloud GPU time for neural-model experiments.

7. Implementation Notes

The script retains the supplied synthetic corpus, rules, classifier, and fusion formula. Small robustness changes add deterministic tie-breaking, parameter validation, configurable display thresholds, and consistent three-decimal score formatting. Substring matching remains intentionally simple: overlapping phrases can count more than once, and generic words can trigger unrelated concepts. These behaviors are limitations of the toy simulator, not evidence of scientific classification quality.

Source: simulator and proposal text supplied by the user. Example output generated by executing the included simulator; no external factual verification was performed.